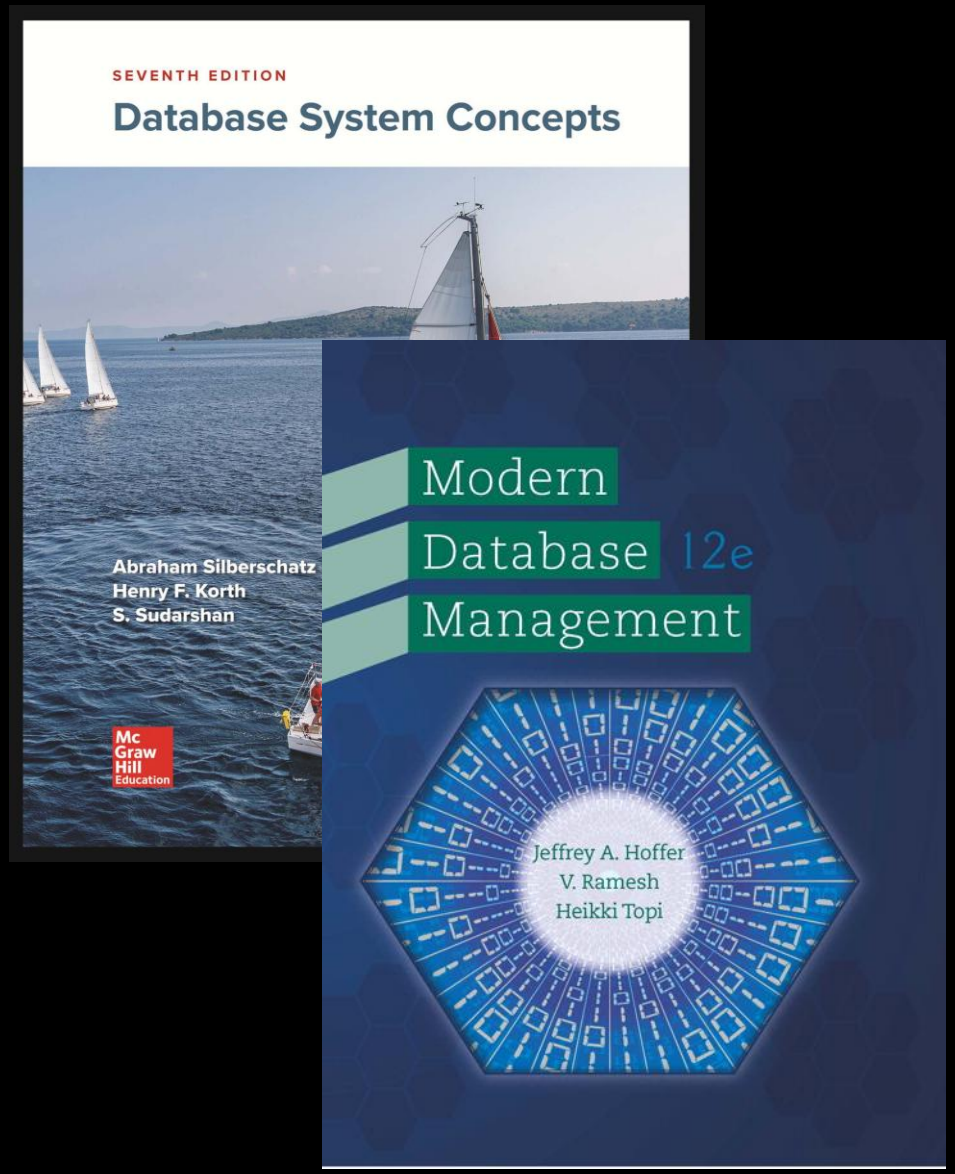


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Introduction to Database

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References

Abraham Silberschatz, Henry F. Korth, S. Sudarshan : "**Database System Concepts**", 7th Edition

- Chapter 1: Introduction

Jeffrey A. Hoffer, Mary B. Prescott, Heikki Topi : "**Modern Database Management**", 12th Edition

- Chapter 1: The Database Environment and Development Process

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Definition (1/3)

Database	organized collection of logically related data
Data	stored representations of meaningful objects and events Structured: numbers, text, dates Unstructured: images, video, documents



Example:

Baker, Kenneth D.	324917628
Doyle, Joan E.	476193248
Finkle, Clive R.	548429344
Lewis, John C.	551742186
McFerran, Debra R.	409723145

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Definition (2/3)

Information	data processed to increase knowledge in the person using the data

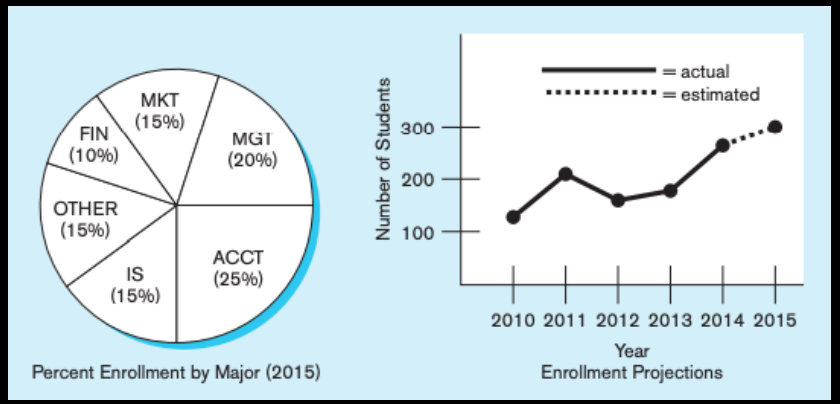
Example:

Class Roster

Course: MGT 500 Business Policy Semester: Spring 2015

Section: 2

Name	ID	Major	GPA
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3



Summarized data - Graphical displays turn data into useful information that managers can use for decision making and interpretation

Data in context - Context helps users understand data

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Definition (3/3)

Metadata	data that describes the properties and context of user data

Example:

TABLE 1-1 Example Metadata for Class Roster

Data Item	Metadata					
Name	**Type**	**Length**	**Min**	**Max**	**Description**	**Source**
Course	Alphanumeric	30			Course ID and name	Academic Unit
Section	Integer	1	1	9	Section number	Registrar
Semester	Alphanumeric	10			Semester and year	Registrar
Name	Alphanumeric	30			Student name	Student IS
ID	Integer	9			Student ID (SSN)	Student IS
Major	Alphanumeric	4			Student major	Student IS
GPA	Decimal	3	0.0	4.0	Student grade point average	Academic Unit

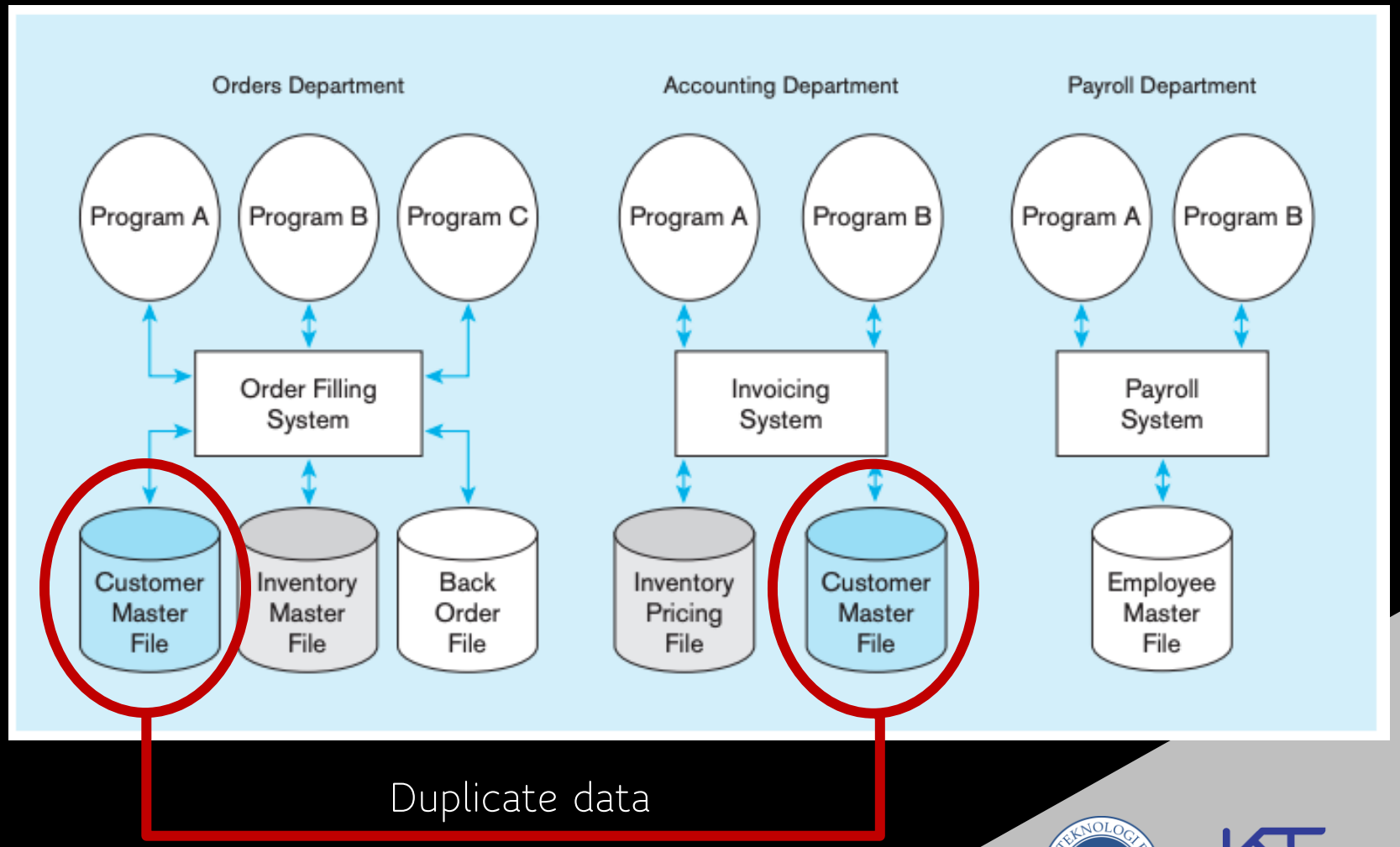
Descriptions of the properties or characteristics of the data, including data types, field sizes, allowable values, and data context

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Traditional File Processing System

When computer-based data processing was first available, there were no databases. To be useful for business applications, computers had to store, manipulate, and retrieve large files of data.

File processing systems at Pine Valley Furniture Company



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Disadvantages of File Processing

Program-Data Dependence	Data Redundancy (Duplication of data)	Limited Data Sharing	Lengthy Development Times	Excessive Program Maintenance
· All programs maintain metadata for each file they use	· Different systems/ programs have separate copies of the same data	· No centralized control of data	· Programmers must design their own file formats	· 80% of information systems budget

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Problems with **Data Dependency**

Each application programmer must maintain their own data

Each application program needs to include code for the metadata of each file

Each application program must have its own processing routines for reading, inserting, updating and deleting data

Lack of coordination and central control

Non-standard file formats

Problems with **Data Redundancy**

Waste of space to have duplicate data

Causes more maintenance headaches

The biggest Problem:

- When data changes in one file, could cause inconsistencies
- Compromises *data integrity*

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Solution : The Database Approach



CENTRAL REPOSITORY OF SHARED DATA



DATA IS MANAGED BY A CONTROLLING AGENT



STORED IN A STANDARDIZED, CONVENIENT FORM

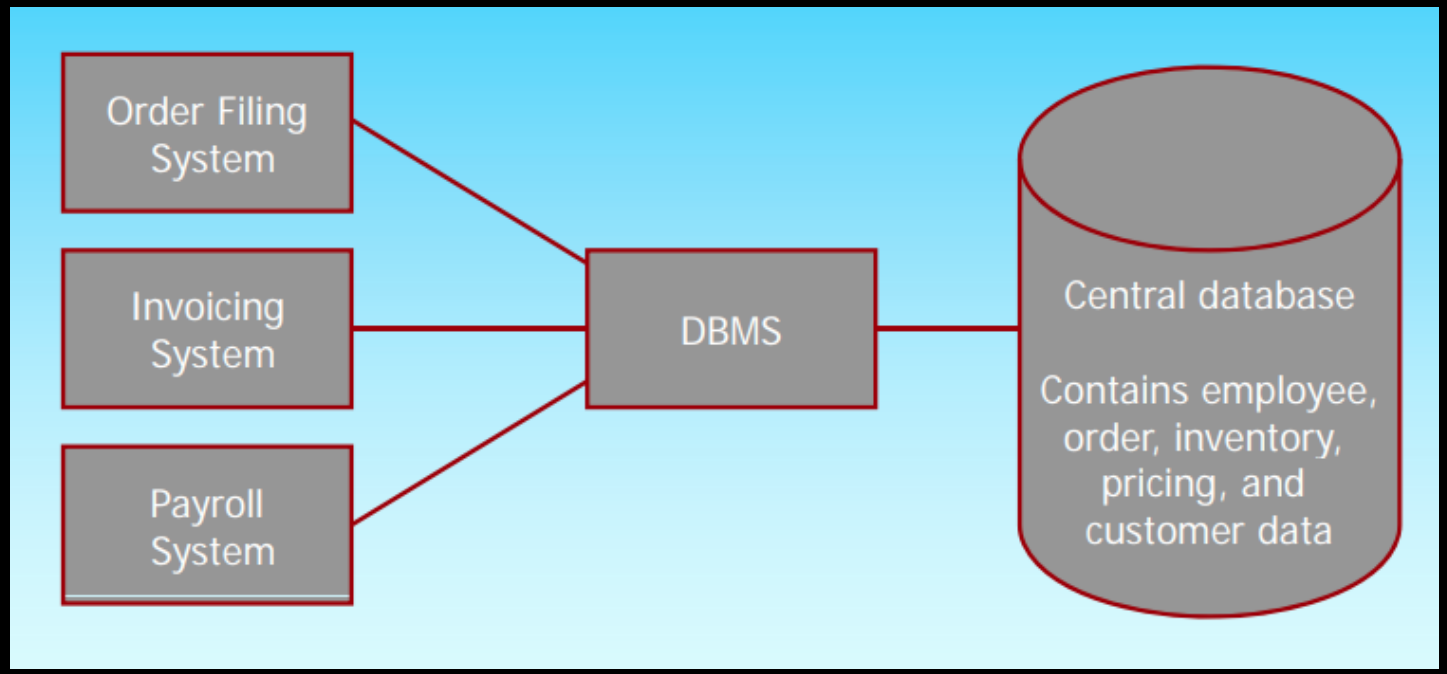
Requires a **Database Management System (DBMS)**

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Database Management System (DBMS)

A software system that is used to create, maintain, and provide controlled access to user databases

DBMS manages data resources like an operating system manages hardware resources



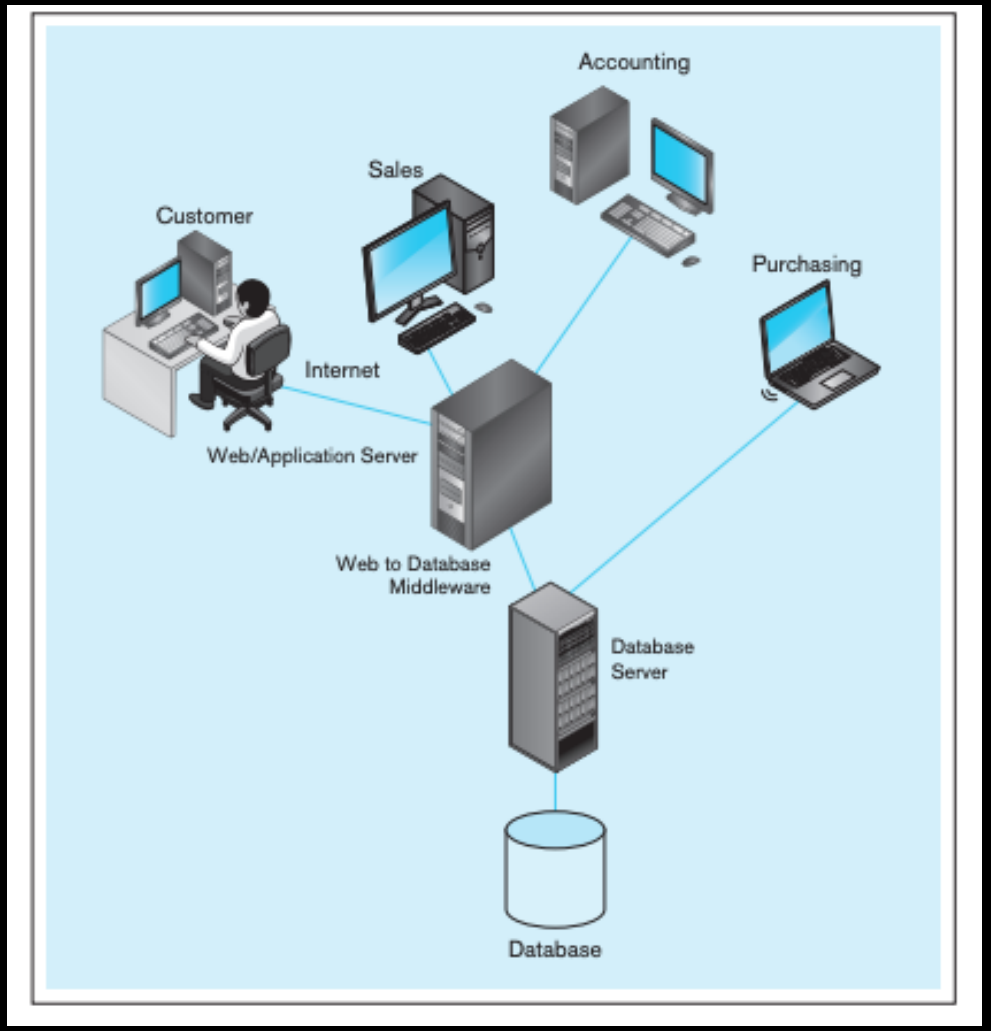
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Advantages of Database Approach

Improved Data Sharing	Enforcement of Standards	Improved Data Quality	Better Data Accessibility/ Responsiveness	Security, Backup/Recovery, Concurrency
· Different users get different views of the data	· All data access is done in the same way	· Constraints, data validation rules	· Use of standard data query language (SQL)	· Disaster recovery is easier

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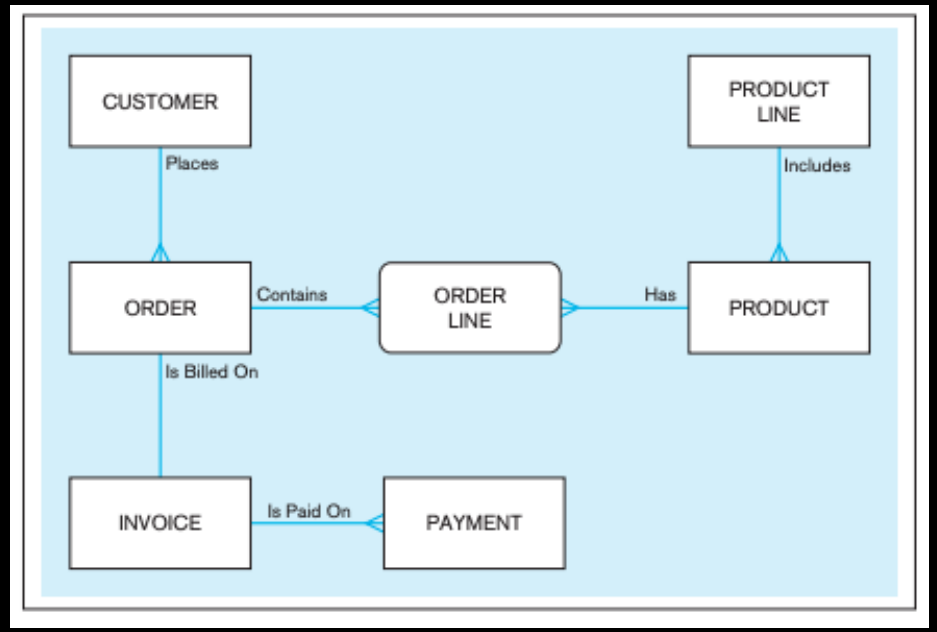
Developing a **database** application for Pine Valley Furniture Company (1/4)



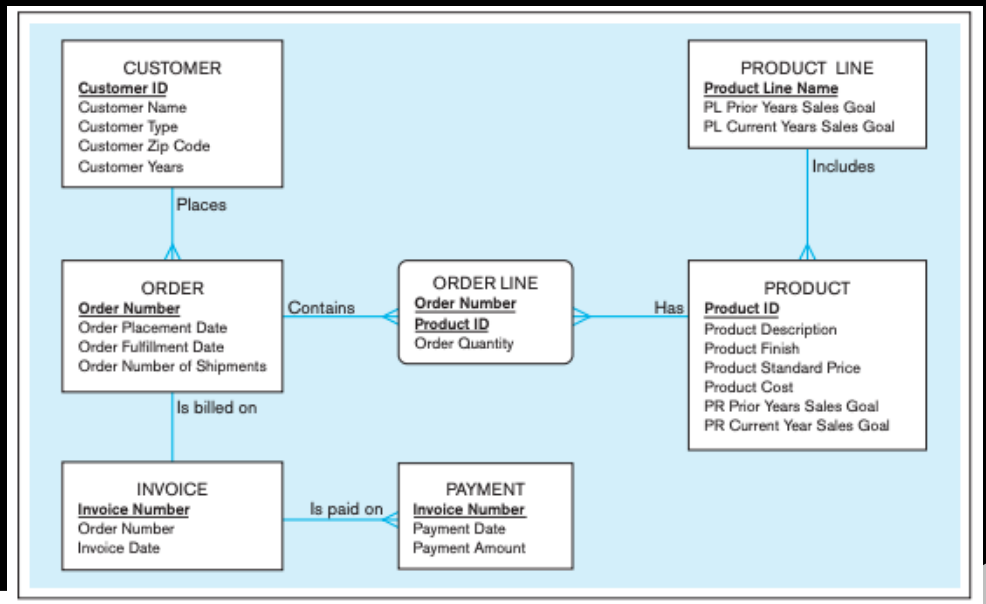
Computer System for Pine Valley Furniture Company

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Developing a **database** application for Pine Valley Furniture Company (2/4)



Preliminary data model for Home Office product line marketing support system

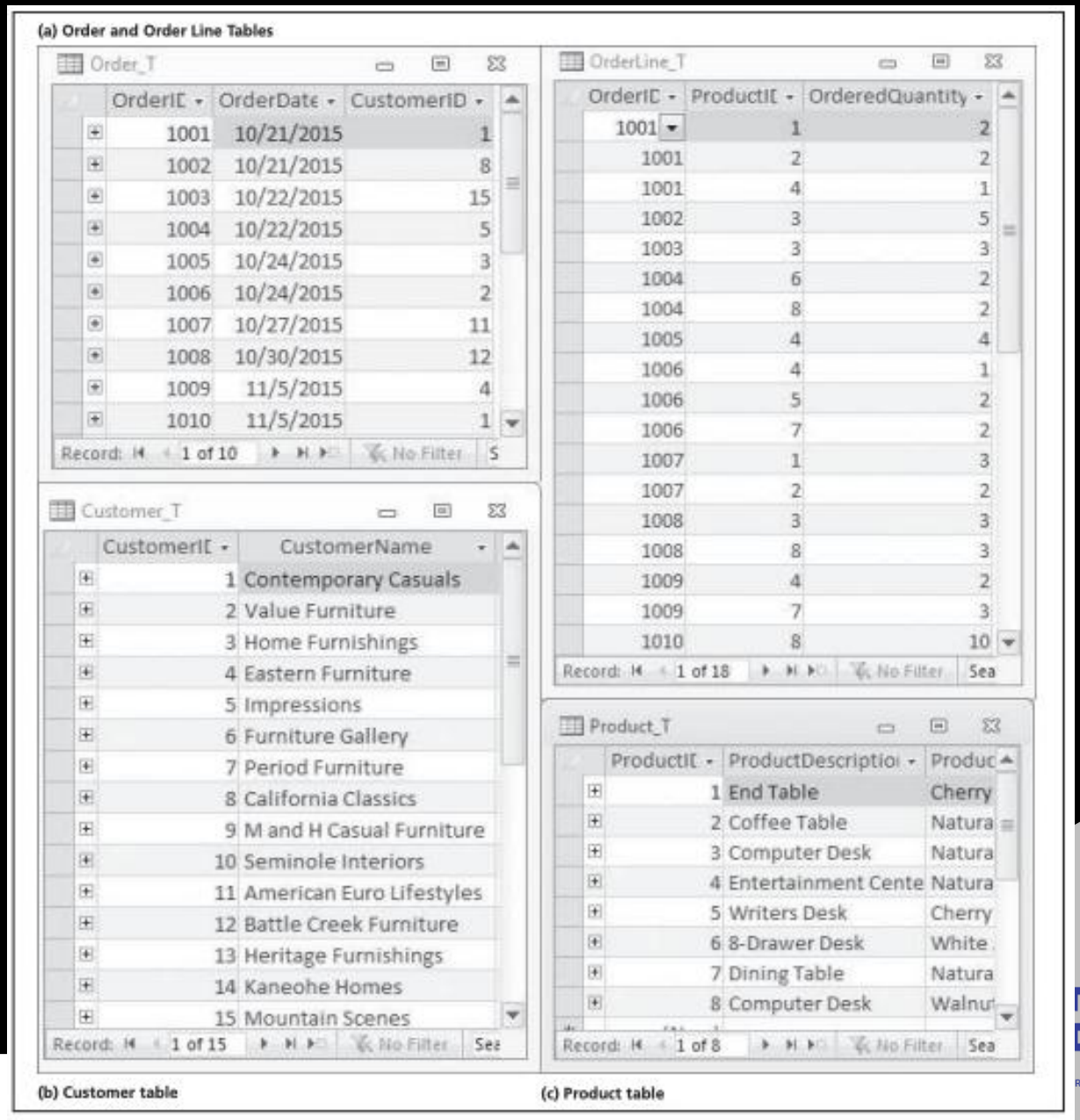


Project data model for Home Office product line marketing support system

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Developing a **database** application for Pine Valley Furniture Company (3/4)

Four **relations** (Pine Valley Furniture Company)



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Developing a **database** application for Pine Valley Furniture Company (4/4)

```
SELECT Product.ProductID, Product.ProductDescription, Product.PRCurrentYearSalesGoal,
       (OrderQuantity * ProductPrice) AS SalesToDate
FROM Order.OrderLine, Product.ProductLine
WHERE Order.OrderNumber = OrderLine.OrderNumber
AND Product.ProductID = OrderedProduct.ProductID
AND Product.ProductID = ProductLine.ProductID
AND Product.ProductLineName = "Home Office";
```

SQL query for Home Office sales-to-goal comparison

Home Office Sales to Date : Select Query

Product ID	Product Description	PR Current Year Sales Goal	Sales to Date
3	Computer Desk	$23,500.00	5625
10	96" Bookcase	$22,500.00	4400
5	Writer's Desk	$26,500.00	650
3	Computer Desk	$23,500.00	3750
7	48" Bookcase	$17,000.00	2250
5	Writer's Desk	$26,500.00	3900

Home Office product line sales comparison

Customer invoice (Pine Valley Furniture Company)

PVFC Customer Invoice

Customer ID	2	Order Number	1006
Customer Name	Value Furniture	Order Date	10/24/2000
Address	15145 S.W. 17th St. Plano, TX 75094		

Product_ID	Product Description	Finish	Quantity	Unit Price	Extended Price:
7	Dining Table	Natural As	2	$800.00	$1,600.00
5	Writer's Desk	Cherry	2	$325.00	$650.00
4	Entertainment Center	Natural M	1	$650.00	$650.00
				Total	$2,900.00

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Costs and Risks of the Database Approach

Up-front costs

- Installation Management Cost and Complexity
- Conversion Costs

Ongoing Costs

- Requires New, Specialized Personnel
- Need for Explicit Backup and Recovery

Organizational Conflict

- Old habits die hard

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